VF-C09 Lace Abrasion

Test Method for Lace Abrasion	**Test method #: VF- C09** **REF #: TM-R-020**
Date Issued: September 15, 2022 **Revision #:2**	**Reason for Revision: Add no failure on abrasion test, performance tensile strength after abrasion**

PURPOSE AND SCOPE: To determine the abrasion resistance of shoelace to repeated rubbing against a standard eyelet, the abrasive action of an eyelet on a standard shoelace, abrasion resistance of a shoelace to repeated rubbing against a similar eyelet, and a lace to carrier abrasion.

Method 1: Abrasion resistance of laces. The lace is rubbed against a standard eyelet.
Reference TM 93
Method 2: Abrasive action of eyelets – The test eyelet is rubbed by standard shoelace.
Reference TM 93
Method 3: Lace to lace abrasion – a second piece of same lace, held open by spacer
Reference TM 154
Method 4: Lace to carrier abrasion - a lace carrier (the eyelet used with the shoe)
Reference TM 154

REFERENCES: SATRA Test Method TM 93, TM 154

EQUIPMENT AND MATERIALS:

I. Lace Abrader which is compliant to SATRA TM 93

1. A means of holding a piece of fiberboard at ang angle of 45±2° to the horizontal. The fiberboard should have an eyelet in its center.

2. A pair of clamps which are capable of holding a shoelace at both ends so that it passes through the eyelet and is bent through 90±5° at the eyelet. One part of the shoelace should be approximately horizontal and the other approximately vertical so that the angle between each part of the shoelace and the fiberboard is 45±5°.See figure 1.

3. A means of holding the portion of the shoelace between the clamp under a constant tension of 500 grams.

4. A means of moving the shoelace with reciprocating motion through a distance of 76±2 mm and back to the original starting position at a rate of 100 ±5 cycles per minute.

5. Counters to record the number of abrasion cycles.

6. Method 1: Abrasion resistance of laces. Four brass eyelets with a barrel of internal dimeter 4.5 mm. These are used as standard eyelets when testing abrasion of shoelaces.

7. Four pieces of round shoelace cotton with core, each of sufficient length to fit between the clamp, and pass through the hole in the fiberboard. These are required for polishing the standard eyelets before testing abrasion resistance of laces, and for assessing the abrasive action of eyelets. When tested with the polished standard eyelets the abrasion resistance should be between 10,000 cycles and 20,000 cycles.

8. Four pieces of rigid fiberboard of thickness 3.0 ±0.5 mm, smallest width at least 20 mm, and suitable size to fit into the machine. The fiberboard has rectangle of either ***38x25mm or 25x25 mm.*** Each piece of fiberboard requires a circular hole drilled or punched in its center, this should be of diameter 5.0 ±0.2 mm if a standard eyelet is being fitted.

9. An eyelet clamp or press with a suitable anvil for clenching the eyelets.

II. Lace Abrader which is compliant to SATRA TM 154

1. A moveable clamp which has a means of firmly holding the following:
    - Method 3 – both ends of a piece of shoelace
    - Method 4- one end of a metal strip.
2. A stationary clamp which is mounted in the same horizontal plane as the moveable clamp and is capable of holding one end of a shoelace. The stationary clamp should be 280±50 mm from the moveable clamp when the clamps are at their minimum separation, this distance will subsequently be referred to as **(D)**
3. A tensioning device with a method of holding the end of a piece of shoelace at a point 35±5 mm to one side of, and in the same horizontal plane as, the stationary clamp and applying a tensioning force of 250 grams. See figure 2. This can be achieved by passing the shoelace over a pulley and suspending a mass of 250±3 grams from the lower end of the vertical portion of the lace.
4. A means of moving the moveable clamp with a simple harmonic reciprocating motion through a distance of 35±2mm and back to the original starting position at a rate of 60±6 cycles per minute.
5. Counters to record the number of abrasion cycles up to failure of the lace or lace carrier
6. For Method 3: lace to lace abrasion.
    - A rigid spacer device of width 25±5 mm. This is used to hold the two legs of the loop of lace which is clamped in the moveable clamp apart, so that when it is under tension the end of the loop forms an isosceles triangle with the spacer as its base and the two parts of the lace are at an angle of 52.5±5.0°. See Figure 2.
    - A template with an angle of 52.5°matked on it. This is for setting the position of the spacer
7. Method 4: lace to carrier abrasion.
    - A metal strip of width approximately 25mm, thickness approximately 3 mm and length approximately **(D)** -75mm. One end of the strip should fit into the moveable clamp and the other end should be capable of holding the test lace carrier.

III. Tension Force: 500 grams based on TM 93 and 250 grams based on TM 154
IV. Scissors
V. Hammer
VI. Hole Punch
VII. Mounting template
VIII. Laces
IX. Eyelets

VF-C09 Lace Abrasion

SAMPLE PREPARATION:

Condition the specimens at 23 ± 2 ℃ temperature and 55 ± 10% relative humidity for 24 hours prior to test.

Note: If sufficient made shoelaces are available, cut each test specimen from a separate lace.

I. Method 1. Abrasion resistance of laces.

1. Cut four pieces of shoelace each of sufficient length to fit between the clamps and pass through the eyelet in the fiberboard mounted on the machine.
2. Each specimen should have a length of 200±10 mm when under a tension of 500 grams.
   Note: It is possible to test specimens with shorter length by tying string to each end to lengthen them, provided only the lace specimen rubs against the eyelet.

II. Method 2. Abrasive action of eyelets.

1. Four eyelets are required as test specimens.

III. Method 3: Lace to lace abrasion:

1. Cut 4 pieces of shoelace each of length 2 x (D-190) mm.
2. Cut 4 pieces of shoelace of length 500 ±10mm

Note: It is possible to test shorter pieces of lace by tying string to each end to lengthen them, provided the string is situated so that only the lace specimen rubs against the lace specimen.

IV. Method 4**:** Lace to carrier abrasion.

1. Use four test lace carriers for the test.
2. Cut four pieces of shoelace each of length 370 ±10 mm.
   Note: It is possible to test shorter pieces of lace by tying string to each end to lengthen them, provided the string is situated so that only the lace specimen rubs against the lace carrier.

REPETITONS: Test four (4) samples

TEST PROCEDURE:

Test should be conducted at standard laboratory conditions @ 23.0±2.0 ˚C temperature and 55 ±10 % Relative Humidity.

Method 1. Abrasion resistance of laces. The lace is rubbed against a standard eyelet.

1. Clench a standard eyelet in the hole in each of the four pieces of fiberboard using the eyelet clamp and mount the pieces of fiberboard into the machine.
2. If the lower side (Point A in figure 1) of any of the eyelets in the machine has been subjected to more than 100,000 abrasion cycles then turn the fiberboard through 180° so that points A and B in figure 1 are reversed.
3. If both sides (Points A and B in Figure 1) of any of the eyelets in the machine have been subjected to more than 100,000 abrasion cycles then discard those eyelets and replace them with new eyelets.
4. Polish any new eyelets in the machine as follows:

4.1 Thread a length of standard lace through each eyelet, and secure one of its ends in each of the clamps.

4.2 Ensure that the portion of lace between the clamps is under a tension of 500 grams and where short lengths of lace are being used, that the eyelets are being rubbed by the lace for the whole cycle.
4.3 Reset all the counters and start the machine.
4.4 Stop the machine after 10,000 rubbing cycles and discard the pieces of standard lace.
4.5 Turn the fiberboard through 180° so that Points A and B in figure 1 are reversed. Repeat the procedure from the beginning of step 4.1 to polish the other side of the eyelet.

5. If any of the pieces of standard lace failed during the polishing procedure then:

   5.1 Discard the eyelet through which the failed lace was threaded
   5.2 Replace the eyelet with new one.
   5.3 Polish the new eyelet with the procedure in step 4.1

6. Thread one of the lace test specimens through each polished eyelet.
7. Secure one end of the lace in each of the clamps.
8. Ensure that the portion of lace between the clamps is under a tension of 500 grams and where short lengths are being used, the eyelets are being rubbed by the lace for the whole cycle.
9. Reset all the counters and start the machine.
10. Stop the test if all the 4 lace test specimens have failed.
11. Record the number of abrasion cycles required to abrade through each of the 4 test specimens.
12. For laces that have not failed the abrasion test, perform a tensile test on the abrasion area to determine the tensile strength after abrasion. Testing should be performed at a speed of 300 mm/min. Record the force at break in kgf.

    NOTE: Refer to Table 1 : Lace abrasion ratings.

Method 2: Abrasive action of eyelets – The test eyelet is rubbed by standard shoelace.

1. Clench a test eyelet in the hole in the center of a piece of fiberboard using the eyelet clamp and place the fiberboard into a test station in the machine with the clenched side lowermost.
2. Repeat the procedure in step 1 for the other three stations of the test machine.
3. Thread a length of the conditioned standard shoelace through each of the test specimen eyelets.
4. Secure the lace and conduct the abrasion test by following the procedure of Method 1 steps 7 to 12.

Method 3: Lace to lace abrasion – a second piece of same lace, held open by spacer

1. Secure both end of one of the shorter pieces of lace in a moveable clamp so that it forms a loop of length approximately 2x(D-190) mm

2. Secure one end of one of the longer pieces of lace in the corresponding stationary clamp. *(Note: It is possible to use pieces of shorter length by trying string to each end to lengthen them, provided only the lace specimen rubs against the lace specimen or lace carrier)*

3. Thread the free end of the piece of lace through the loop formed by the piece of lace.

4. Apply a tensioning force of 250 grams to the free end of the piece of lace.
5. Fit a spacer into the loop formed by the piece of lace and slide it towards the point of contact with the piece of lace until the template indicates that the tip of the loop of the lace forms an angle of 52.5 ±5.0°. See figure 2

6. Repeat the procedure from step 1-5 for the other 3 pairs of test laces.

7. Reset all the counters and start the machine

8. Stop the test when all the 4 lace specimens have failed.

9. For each lace test specimen record the number of cycles to failure, and the type of failure as failure of the shorter lace or failure of the longer lace. Also record a description of the respective amount of wear in the covering and core of each lace.

10. For laces that have not failed the abrasion test, perform a tensile test on the abrasion area to determine the tensile strength after abrasion. Testing should be performed at a speed of 300 mm/min. Record the force at break in kgf.

    NOTE: Refer to Table 1: Lace abrasion ratings.

Method 4: Lace to carrier abrasion - a lace carrier (the eyelet used with the shoe)

1. Fix a test lace carrier to one end of each of the metal strips so that the relative orientation between the carrier and the metal strip is the same as the orientation between the carrier and the facing in the completed shoe.

2. Clamp the other end of each metal strip into each of the moveable clamps so that the lace carrier is approximately (D-75) mm from the moveable clamp.

3. Secure one end of one of the pieces of lace in the corresponding stationary clamp.

(Note: It is possible to use pieces of shorter length by trying string to each end to lengthen them, provided only the lace specimen rubs against the lace specimen or lace carrier)

4. Thread the free end of the pieces of lace through the lace carrier.

5. Apply a tensioning force of 250 grams to the free end of the piece of lace. See figure 3.

6. Repeat the procedure from step1-5 for the other 3 test samples.

7. Reset the counters and start the machine.

8. Stop the test when all four specimens have failed.

9. For each test specimen record the number of cycles to failure, and the type of failure as failure of the lace or failure of the carrier. Also record a description of the respective amount of wear in the covering and core of each lace and any damage to the carrier.

10. For laces that have not failed the abrasion test, perform a tensile test on the abrasion area to determine the tensile strength after abrasion. Testing should be performed at a speed of 300 mm/min. Record the force at break in kgf.

NOTE: Refer to Table 1 : Lace abrasion ratings.

Report:

Report the lace description, the number of cycles to failure, abrasion rating, the type of failure, force at break before and after abrasion, and any deviations from this procedure.

Table 1. Lace Abrasion Ratings

Rating	Description
1	No Visible Wear
2	Slight dulling or matting of top surface with no pilling, or fraying
3	Up to 50% wear through top surface with visible pilling, or fraying
4	50% to 100% wear through top surface
5	Lace broken in two

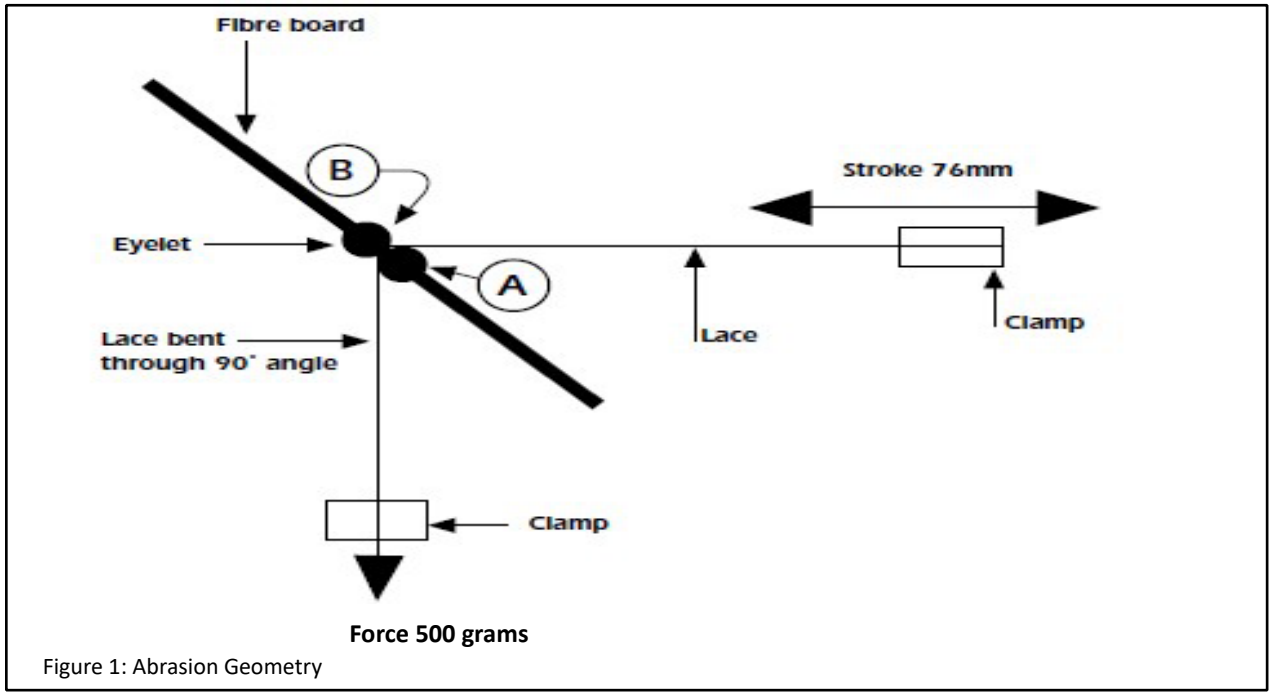


Figure 1: Abrasion Geometry

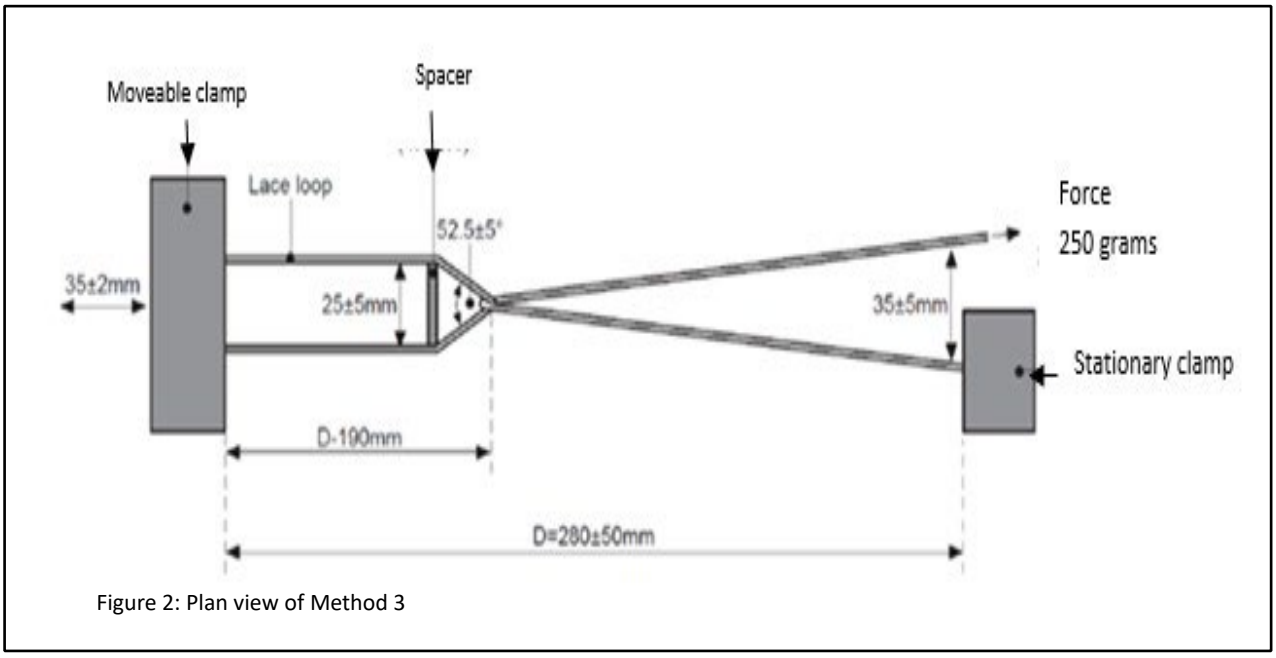


Figure 2: Plan view of Method 3

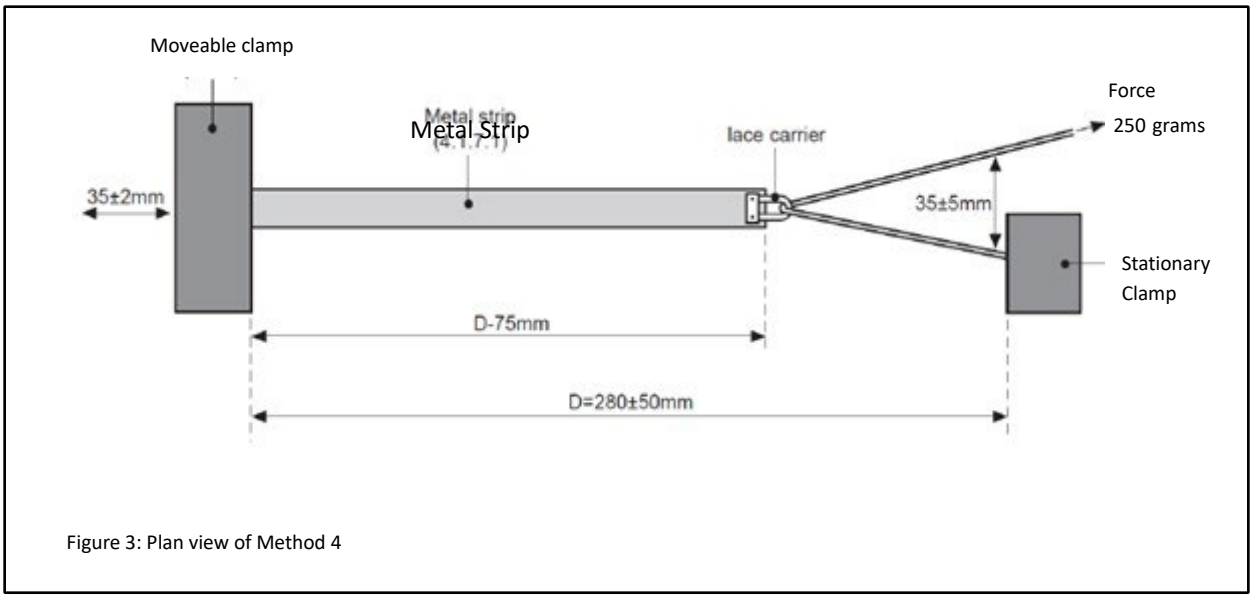


Figure 3: Plan view of Method 4

Sample of Machines	
Photo	**Photo**
	
Test Method 1&2	**Test Method 3&4**
Machine Name : GT-7012-E1	**Machine Name : GT-7012-E**
Reference # : TM 93	**Reference # : TM 154**
Tension Force : 500 grams	**Tension Force : 250 grams**